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Descriptive Statistics

Dr. Govindu

Stem-and-Leaf Graphs

- One simple graph, the stem-and-leaf graph or stem-plot, comes from the field of exploratory data analysis. It is a good choice when the data sets are small.
- To create the plot, divide each observation of data into a stem and a leaf. The leaf consists of a final significant digit.

Stem	Leaf
3	3
4	2 9 9
5	3 5 5
6	1 3 7 8 8 9 9
7	2 3 4 8
8	0 3 8 8 8
9	0 2 4 4 4 4 6
10	0

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Stem-and-Leaf Graphs

- For Susan Dean's spring pre-calculus class, scores for the first exam were as follows (smallest to largest): 33; 42; 49; 49; 53; 55; 55; 61; 63; 67; 68; 68; 69; 69; 72; 73; 74; 78; 80; 83; 88; 88; 88; 90; 92; 94; 94; 94; 94; 96; 100

The stem-plot shows that most scores fell in the 60s, 70s, 80s, and 90s. Eight out of the 31 scores or approximately 26% were in the 90s or 100, a fairly high number of As.

Stem	Leaf
3	3
4	2 9 9
5	3 5 5
6	1 3 7 8 8 9 9
7	2 3 4 8
8	0 3 8 8 8
9	0 2 4 4 4 4 6
10	0

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Stem-and-Leaf Graphs

- One simple graph, the stem-and-leaf graph or stem-plot, comes from the field of exploratory data analysis. It is a good choice when the data sets are small.
- To create the plot, divide each observation of data into a stem and a leaf. The leaf consists of a final significant digit.

Stem	Leaf
3	3
4	2 9 9
5	3 5 5
6	1 3 7 8 8 9 9
7	2 3 4 8
8	0 3 8 8 8
9	0 2 4 4 4 4 6
10	0

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Example

- The data are the distances (in kilometers) from a home to local supermarkets. Create a stem-plot using the data: 1.1; 1.5; 2.3; 2.5; 2.7; 3.2; 3.3; 3.3; 3.5; 3.8; 4.0; 4.2; 4.5; 4.5; 4.7; 4.8; 5.5; 5.6; 6.5; 6.7; 12.3

Do the data seem to have any concentration of values?

The value 12.3 may be an outlier. Values appear to concentrate at three and four kilometers.

Stem	Leaf
1	1 5
2	3 5 7
3	2 3 3 5 8
4	0 2 5 5 7 8
5	5 6
6	5 7
7	
8	
9	
10	
11	
12	3

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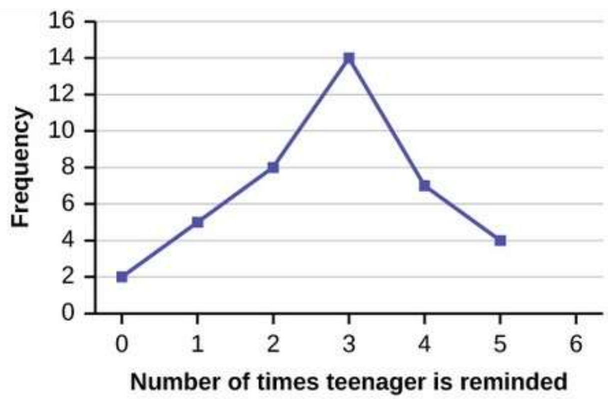
Line Graphs

- Another type of graph that is useful for specific data values is a line graph.

Example

- In a survey, 40 mothers were asked how many times per week a teenager must be reminded to do his or her chores. The results are shown in Table and in Figure.

Number of times teenager is reminded	Frequency
0	2
1	5
2	8
3	14
4	7
5	4



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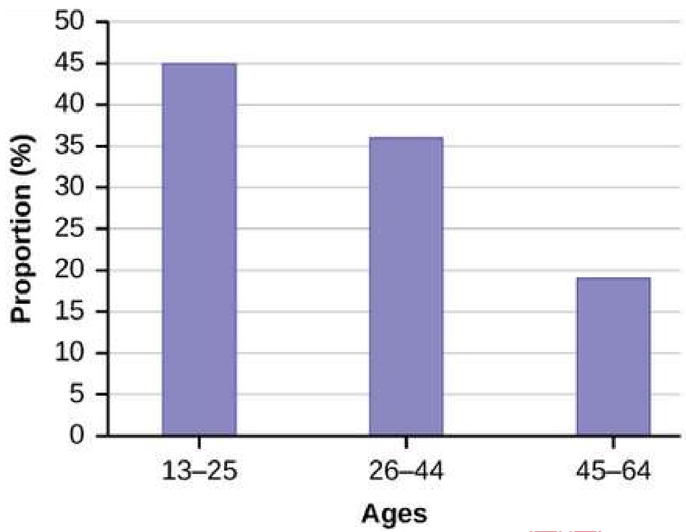
Bar graphs

- Bar graphs consist of bars that are separated from each other. The bars can be rectangles or they can be rectangular boxes (used in three-dimensional plots), and they can be vertical or horizontal.

- **Example**

- By the end of 2011, Facebook had over 146 million users in the United States. Table shows three age groups, the number of users in each age group, and the proportion (%) of users in each age group. Construct a bar graph using this data.

Age groups	Number of Facebook users	Proporti on
13–25	65,082,280	45%
26–44	53,300,200	36%
45–64	27,885,100	19%



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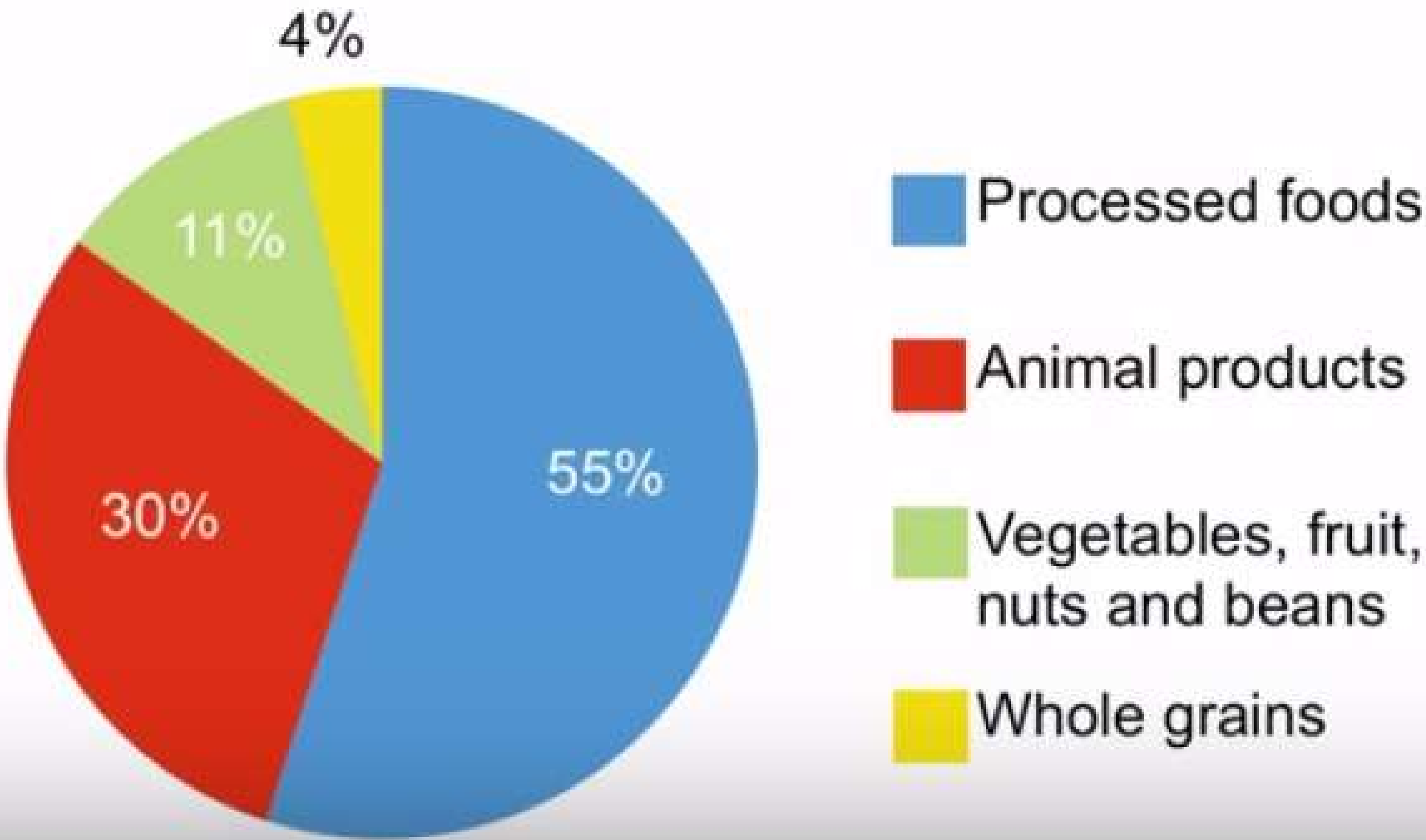
The Standard American Diet (SAD):

promotes chronic disease and weakens immune function



Histogram

- A histogram consists of contiguous (adjoining) boxes. It has both a horizontal axis and a vertical axis. The horizontal axis is labeled with what the data represents (for instance, distance from your home to school). The vertical axis is labeled either frequency or relative frequency (or percent frequency or probability). The graph will have the same shape with either label. The histogram (like the stem-plot) can give you the shape of the data, the center, and the spread of the data. A rule of thumb is to use a histogram when the data set consists of 100 values or more.

Histogram

- To construct a histogram, first decide how many bars or intervals, also called classes, represent the data. Many histograms consist of five to 15 bars or classes for clarity. The number of bars needs to be chosen.
- Choose a starting point for the first interval to be less than the smallest data value. A convenient starting point is a lower value carried out to one more decimal place than the value with the most decimal places.

Example

- The following data are the heights (in inches to the nearest half inch) of 100 male semiprofessional soccer players. The heights are continuous data, since height is measured.
- 60; 60.5; 61; 61; 61.5; 63.5; 63.5; 63.5; 64; 64; 64; 64; 64; 64; 64; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 68; 68; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69.5; 69.5; 69.5; 69.5; 69.5; 70; 70; 70; 70; 70; 70; 70.5; 70.5; 70.5; 71; 71; 71; 72; 72; 72; 72.5; 72.5; 73; 73.5; 74

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Example

- The smallest data value is 60. Since the data with the most decimal places has one decimal (for instance, 61.5), we want our starting point to have two decimal places. Since the numbers 0.5, 0.05, 0.005, etc. are convenient numbers, use 0.05 and subtract it from 60, the smallest value, for the convenient starting point.

- 60 – 0.05 = 59.95 which is more precise than, say, 61.5 by one decimal place.
- The starting point is, then, 59.95.
- The largest value is 74, so 74 + 0.05 = 74.05 is the ending value.

Example

- Next, calculate the width of each bar or class interval. To calculate this width, subtract the starting point from the ending value and divide by the number of bars (you must choose the number of bars you desire). For example, if there are n values of data, take the square root of n and round to next integer. Suppose you choose eight bars.

$$\frac{74.05-59.95}{8}=1.76$$

- We will round up to two and make each bar or class interval two units wide.

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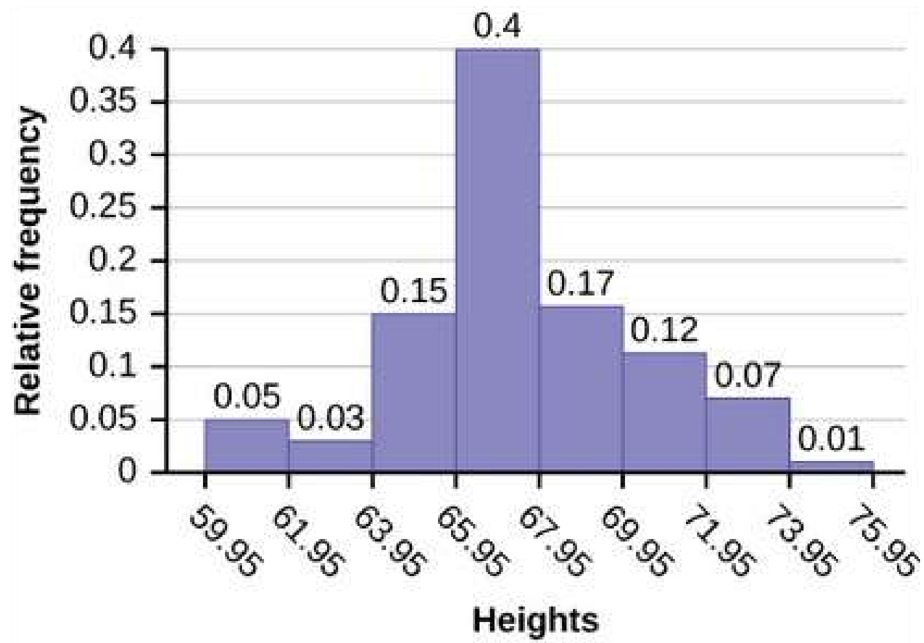
Example

- We will round up to two and make each bar or class interval two units wide.

The boundaries are:

- 59.95
- 59.95 + 2 = 61.95
- 61.95 + 2 = 63.95
- 63.95 + 2 = 65.95
- 65.95 + 2 = 67.95
- 67.95 + 2 = 69.95
- 69.95 + 2 = 71.95
- 71.95 + 2 = 73.95
- 73.95 + 2 = 75.95



60; 60.5; 61; 61; 61.5; 63.5; 63.5; 63.5; 64; 64; 64; 64; 64; 64; 64; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 64.5; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 66.5; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 67.5; 68; 68; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69; 69.5; 69.5; 69.5; 69.5; 69.5; 70; 70; 70; 70; 70; 70; 70.5; 70.5; 70.5; 71; 71; 71; 72; 72; 72; 72.5; 72.5; 73; 73.5; 74

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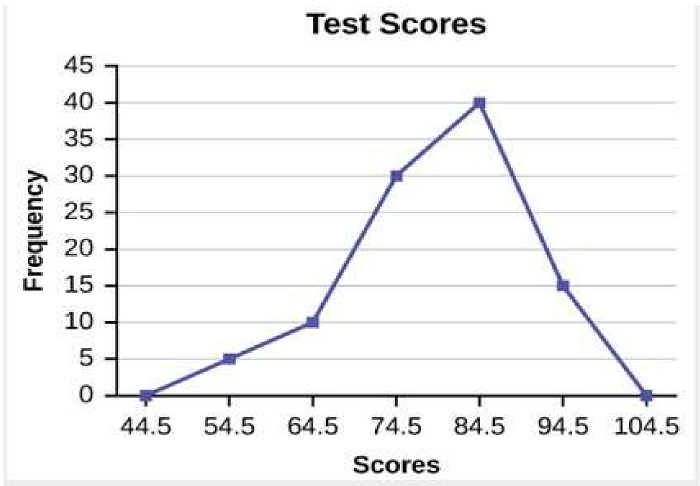
Frequency polygons

- Frequency polygons are analogous to line graphs, and just as line graphs make continuous data visually easy to interpret, so too do frequency polygons.
- To construct a frequency polygon, first organize the data as if you are graphing a histogram. Then, mark the midpoints of the intervals on the x-axis. Now, plot the (the midpoint, frequency) of each interval. Connect the points like you do with a line graph. Also, plot points one interval below the lowest midpoint with zero frequency, and one interval above the highest midpoint with zero frequency; note that these point are included to force the graph touch the x-axis to give it the closed polygon shape.

Example

- **Frequency Distribution for Calculus Final Test Scores:**

Lower Bound	Upper Bound	Frequency	Cumulative Frequency
49.5	59.5	5	5
59.5	69.5	10	15
69.5	79.5	30	45
79.5	89.5	40	85
89.5	99.5	15	100



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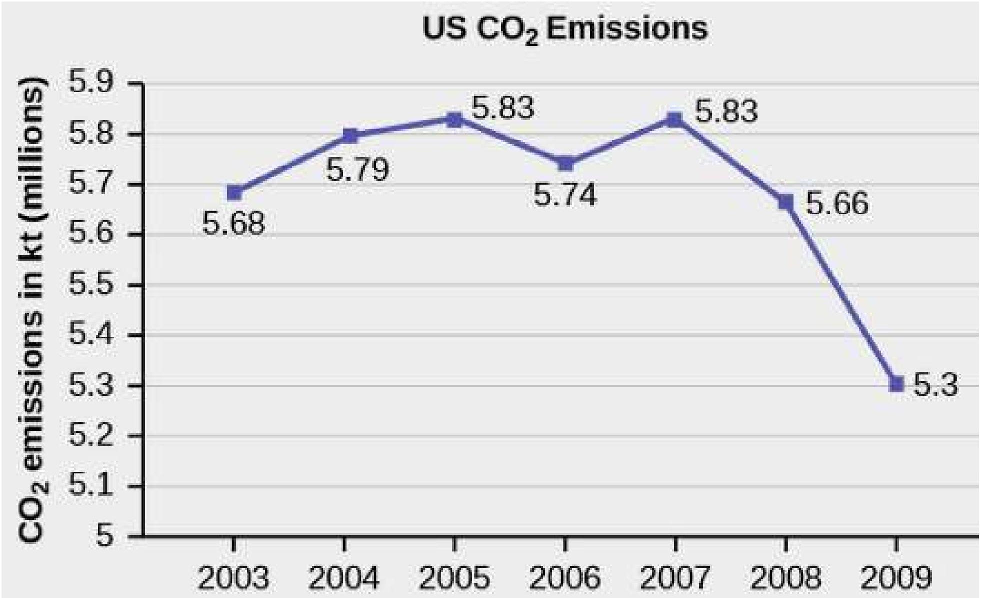
Time Series Graphs

- To construct a time series graph, we must look at both pieces of our paired data set. We start with a standard Cartesian coordinate system. The horizontal axis is used to plot the date or time increments, and the vertical axis is used to plot the values of the variable that we are measuring. By doing this, we make each point on the graph correspond to a date and a measured quantity. The points on the graph are typically connected by straight lines in the order in which they occur.

Example

- The following table is a portion of a data set from www.worldbank.org. Use the table to construct a time series graph for CO2 emissions for the United States.

United States	CO2 Emissions
2003	5,681,664
2004	5,790,761
2005	5,826,394
2006	5,737,615
2007	5,828,697
2008	5,656,839
2009	5,299,563



Percentile and Quartiles

- To calculate quartiles and percentiles, the data must be ordered from smallest to largest. Quartiles divide ordered data into quarters. Percentiles divide ordered data into hundredths. To score in the 90th percentile of an exam does not mean, necessarily, that you received 90% on a test. It means that 90% of test scores are the same or less than your score and 10% of the test scores are the same or greater than your test score.

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Percentile and Quartiles

- **Formulas for Finding the kth Percentile and kth Quartile:**

$$i = \frac{k}{100}(n+1)$$

$$i = \frac{k}{4}(n+1)$$

Example

- Listed are 29 ages for Academy Award winning best actors in order from smallest to largest.
- 18; 21; 22; 25; 26; 27; 29; 30; 31; 33; 36; 37; 41; 42; 47; 52; 55; 57; 58; 62; 64; 67; 69; 71; 72; 73; 74; 76; 77
- Find the 70th percentile.
- Find the 83rd percentile.

Example

$$i = \frac{k}{100}(n + 1) = \frac{70}{100}(29+1) = 21.$$

Twenty-one is an integer, and the data value in the 21st position in the ordered data set is 64. The 70th percentile is 64 years.

$$i = \frac{k}{100}(n+1) = \frac{83}{100}(29+1) = 24.9.$$

24.9 is NOT an integer. Round it down to 24 and up to 25. The age in the 24th position is 71 and the age in the 25th position is 72. Average 71 and 72. The 83rd percentile is 71.5 years.

Example

- For the following 13 real estate prices, calculate the IQR and determine if any prices are potential outliers. Prices are in dollars.
- 389,950; 230,500; 158,000; 479,000; 639,000; 114,950; 5,500,000; 387,000; 659,000; 529,000; 575,000; 488,800; 1,095,000
- Order the data from smallest to largest. The IQR is the distance between Q1 and Q3.
- 114,950; 158,000; 230,500; 387,000; 389,950; 479,000; 488,800; 529,000; 575,000; 639,000; 659,000; 1,095,000; 5,500,000

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Example

$$i = \frac{k}{4}(n + 1) = \frac{1}{4}(13+1) = 3.5,$$

so, round down and up, and take the average of 3rd and 4th:

$$Q_1 = \frac{230,500+387,000}{2} = 308,750$$

$$i = \frac{k}{4}(n + 1) = \frac{3}{4}(13+1) = 10.5,$$

so, round down and up, and take the average of 10th and 11th:

$$Q_3 = \frac{639,000+659,000}{2} = 649,000$$

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Example

- Order the data from smallest to largest. The IQR is the distance between Q1 and Q3.
- 114,950; 158,000; 230,500; 387,000; 389,950; 479,000; 488,800; 529,000; 575,000; 639,000; 659,000; 1,095,000; 5,500,000

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so, round down and up, and take the average of 10th and 11th:

$$Q_3 = \frac{639,000 + 659,000}{2} = 649,000$$

Example

- IQR = 649,000 – 308,750 = 340,250

- (1.5)(IQR) = (1.5)(340,250) = 510,375

- Q_1 – (1.5)(IQR) = 308,750 – 510,375 = –201,625
- Q_3 + (1.5)(IQR) = 649,000 + 510,375 = 1,159,375

- No house price is less than –201,625. However, 5,500,000 is more than 1,159,375. Therefore, 5,500,000 is a potential outlier.

Example

- A Formula for Finding the Percentile of a Value in a Data Set:
  Order the data from smallest to largest.

  Calculate $\frac{x+0.5y}{n}(100)$, then round to the nearest integer.
- Where:
- x = the number of data values counting from the bottom of the data list up to but not including the data value for which you want to find the percentile.
- y = the number of data values equal to the data value for which you want to find the percentile.
- n = the total number of data.

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Example

- Listed are 29 ages for Academy Award winning best actors in order from smallest to largest.
- 18; 21; 22; 25; 26; 27; 29; 30; 31; 33; 36; 37; 41; 42; 47; 52; 55; 57; 58; 62; 64; 67; 69; 71; 72; 73; 74; 76; 77
- Find the percentile for 58.
- Find the percentile for 25.

Example

Counting from the bottom of the list, there are 18 *data values less than* 58. *There is one value of* 58, *so for* $x = 18$ *and* $y = 1$:.

$$\frac{x+0.5y}{n}(100) = \frac{18+0.5(1)}{29}(100) = 63.8.$$

So, 58 *is the* 64*th percentile.*

Counting from the bottom of the list, there are 3 *data values less than* 25. *There is one value of* 25, *so for* $x = 3$ *and* $y = 1$:.

$$\frac{x+0.5y}{n}(100) = \frac{3+0.5(1)}{29}(100) = 12.07.$$

So, 25 *is the* 12*th percentile.*

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Interpreting Percentiles, Quartiles, and Median

- Median is the 50th Percentile or the 2nd Quartile. A percentile may or may not correspond to a value judgment about whether it is “good” or “bad.” The interpretation of whether a certain percentile is “good” or “bad” depends on the context of the situation to which the data applies. In some situations, a low percentile would be considered “good;” in other contexts a high percentile might be considered "good". In many situations, there is no value judgment that applies.

Example

- On a timed math test, the first quartile for time it took to finish the exam was 35 minutes. Interpret the first quartile in the context of this situation.
- Twenty-five percent of students finished the exam in 35 minutes or less.
- Seventy-five percent of students finished the exam in 35 minutes or more.
- A low percentile could be considered good, as finishing more quickly on a timed exam is desirable. (If you take too long, you might not be able to finish.)

Example

- At a community college, it was found that the 30th percentile of credit units that students are enrolled for is seven units. Interpret the 30th percentile in the context of this situation.
- Thirty percent of students are enrolled in seven or fewer credit units.
- Seventy percent of students are enrolled in seven or more credit units.
- In this example, there is no “good” or “bad” value judgment associated with a higher or lower percentile. Students attend community college for varied reasons and needs, and their course load varies according to their needs.